Technology Stack

Date	04 July 2026
Team ID	
Project Name	Rising Waters
Maximum Marks	2 Marks

Define Your Technology Stack:

Identify the programming languages, frameworks, databases, front-end tools, back-end tools, and APIs that you will use to build your solution. A well-chosen technology stack ensures that your application is scalable, maintainable, and aligned with your project requirements.

Fill out the table below to detail the specific technologies chosen for each component of your architecture and provide a brief justification for your choice.

Technology Stack Details

Component	Technology Chosen	Justification
Programming Language	Python	Easy to develop machine learning models and process data efficiently.
Machine Learning Framework	Scikit-learn	Provides reliable algorithms for training and evaluating flood prediction models.
Data Analysis Libraries	Pandas, NumPy	Used for data cleaning, preprocessing, and numerical computations.
Data Visualization	Matplotlib, Plotly	Creates graphs and charts for Exploratory Data Analysis (EDA).

Component	Technology Chosen	Justification
Front-End	HTML5, CSS3, JavaScript, Bootstrap	Builds a responsive, user-friendly web interface.
Back-End	Flask	Lightweight Python framework for integrating the ML model with the web application.
Database	SQLite (or MySQL)	Stores user information, prediction history, and application data.
Model Storage	Pickle (.pkl)	Saves the trained machine learning model for prediction.
Development Environment	Visual Studio Code, Jupyter Notebook	Used for coding, testing, and model development.
Version Control	Git & GitHub	Tracks code changes and supports team collaboration.
Deployment Platform	Render / Railway / Heroku	Hosts the web application online for user access.
APIs (Optional)	OpenWeather API	Retrieves real-time weather data to improve flood predictions.
Operating System	Windows / Linux	Supports development and deployment of the application.